



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Project Task 3

Programme: Bachelor of Computer Science (Data Engineering)

Subject: SECR1213 Network Communications

Session-Sem: 2024/2025-1

Section: 02

Group name: Router ranger

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Index

3.1 Introduction list of devices uses	3
3.2 Table of Devices	6
3.3 Reflections	11
3.3.1 Are you surprised by the prices? How were you surprised?	11
3.3.2 Have you ever considered cost as a factor for choosing networking devices?	11
3.3.3 What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.	12
Meeting Minutes	18
Meeting Minutes #1	18
Meeting Minutes #2	20
Reference	22

3.1 Introduction list of devices uses

1. Routers

A router is a device designed to link two or more packet-switched networks or subnetworks. Its main roles include directing data packets to their correct IP addresses and enabling multiple devices to share a single Internet connection [1]. While there are various types of routers, most are used to transmit data between local area networks (LANs) and wide area networks (WANs). A LAN refers to a network of interconnected devices within a limited geographical area and typically requires only one router [1]. The key functions of a router include network traffic management and IP address assignment. Firstly, a router manages network traffic by routing data between various devices, such as computers, smartphones, and printers, within a local network as well as between the local network and the internet. Secondly, it handles IP address assignment by using the Dynamic Host Configuration Protocol (DHCP) to assign unique IP addresses to each device on the network, ensuring smooth and efficient communication.

2. Switches

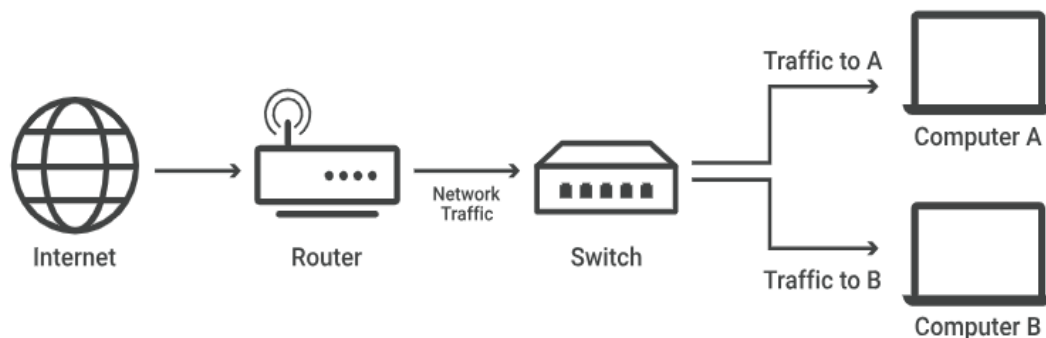


Figure 3.1.1 Switches

A network switch links devices within a network, allowing them to communicate by exchanging data packets [2]. Switches connect various devices, including computers, wireless access points, printers, and servers, within a building or campus. They are crucial parts of any network. The switch facilitates information sharing and effective communication between these linked devices [3].

3. Patch panels

A patch panel is a networking device that has several ports for connecting different Ethernet cables [4]. It simplifies cable management by offering a single spot for classifying and marking every port [4]. Additionally, patch panels facilitate the separation of several networks and enable quick troubleshooting of any difficulties that may develop. For instance, fiber -optic patch panels are widely used patch panel

types that are perfect for supporting a variety of optical fiber connections since they are specialized in using light signals to transmit data and images over thin optical fiber [4].

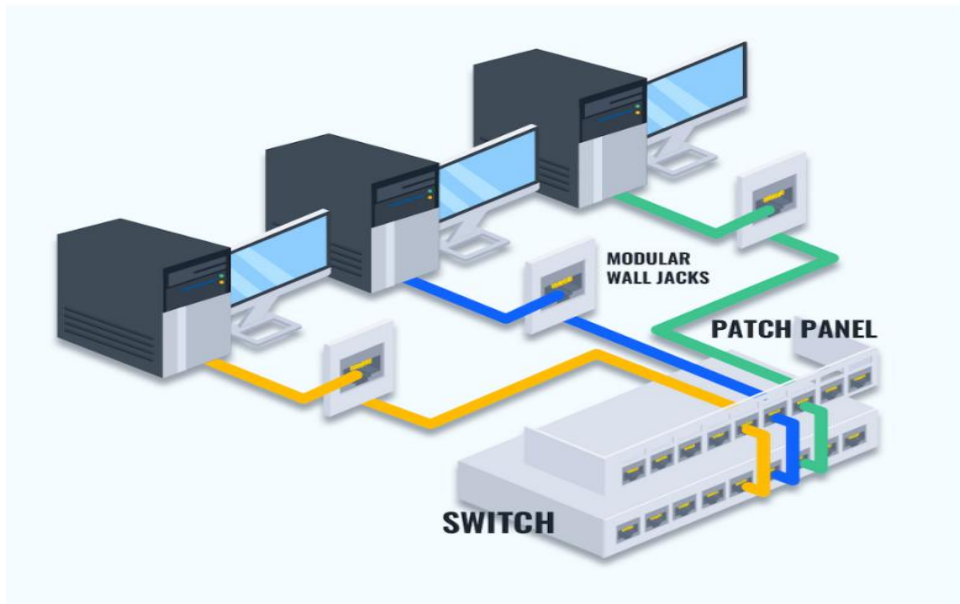


Figure 3.1.2

4. Ethernet cables

An Ethernet cable is a type of networking cable that connects computers, routers, switches, and other network devices to a local area network (LAN) or the Internet. It is mostly utilized for wired data transmission, providing a steady and dependable connection for fast data transfers. Multiple twisted pairs of copper wires are commonly seen in Ethernet cables, which serve to increase signal quality and lower interference. varying types of Ethernet cables (such as Cat5e, Cat6, Cat6a, Cat7, and Cat8) have varying frequencies and maximum data transfer speeds. Cables in higher categories (such as Cat6 and above) can travel farther and quicker.

5. Wireless access point

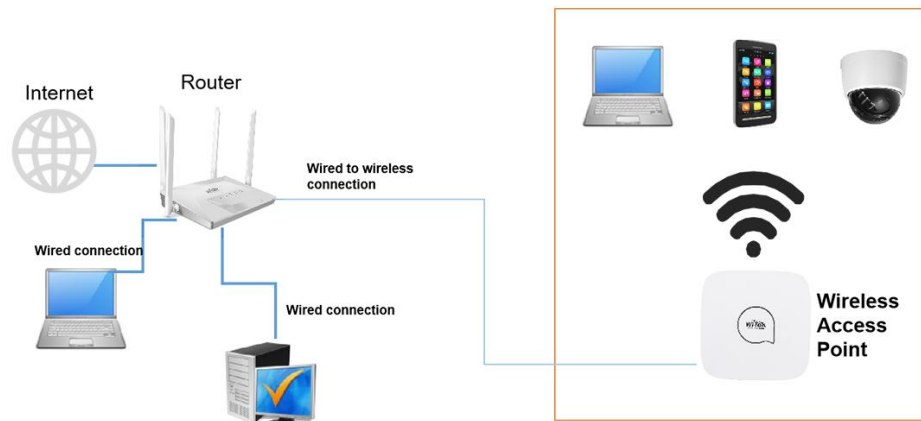


Figure 3.1.3

Wireless access point is a networking device that connects devices with wireless-networking capabilities to access the wired network given that they must be within connection range of the wireless AP. This includes routers, switches, endpoint devices etc. Wireless AP can connect the clients to the network using specific frequencies. However, the range covered can be influenced by factors such as its placement, surrounding obstacles, active devices frequency disruption and others, hence wireless AP can also function as a signal range extender to provide wider wireless coverage [5]. What makes wireless AP stands out from wired AP is that its more convenient to connect all the network devices to the network wirelessly rather than using physical wires and cables.

6. Fiber Optics

Fiber optics is a recent technology that act as a medium to allow data transmission in the form of light pulses over long distances at a high speed. It's primarily made from silica glass for better refraction of light. What makes fiber optics stand out from coaxial Internet cables is its reliability during power outage or electricity interruption. As fiber optics uses light to transmit data instead of electricity, its operational function won't be disrupted hence making it more favourable in these situations [6]. Fiber optic Internet promotes higher bandwidth and can reach upload and download speeds of a range of at least 250 to 1,000 Mbps [6]. It has significantly improved the Internet connection speed, making it more reliable in high density urban areas or working environments where fast and reliable connection is substantial.

3.2 Table of Devices

Devices	Description	Price per unit (RM)	Quantity	Total price (RM)
Router  Figure 3.2.1	TP-Link Archer AX90 (AX6600) tri-band Wi-Fi 6 setup for fast, reliable wireless connectivity. It supports speeds up to 4804 Mbps on 5 GHz for high-speed applications, 1201 Mbps on a secondary 5 GHz band, and 574 Mbps on 2.4 GHz, providing flexibility for various usage scenarios Ultra Connectivity: Equipped with one 2.5 Gbps WAN/LAN port and one 1 Gbps WAN/LAN port, along with three additional Gigabit LAN ports and two USB ports for media sharing and storage, it maximizes connectivity options Coverage and Performance: Features eight high-gain antennas with Beamforming technology to ensure extended coverage, while OFDMA and MU-MIMO allow for low-latency connections to multiple devices simultaneously Processor: Powered by a 1.5 GHz quad-core processor, the Archer AX90 ensures fast data processing for smooth operation in busy networks	799.00	29	23171
Switches  Figure 3.2.2	TP-Link TL-SG3428X High-Speed Connectivity: The switch has 24 Gigabit RJ45 ports that ensure reliable, high-speed connections for connected devices, which is essential for smooth data flow in a building environment	1445.00	24	34680

	10G Uplink Ports: It includes 4 × 10 Gbps SFP+ slots for uplinks, allowing high-bandwidth connections to the core network or other switches. This is critical for avoiding bottlenecks, especially in large networks with heavy traffic Advanced Security: Offers IP-MAC-Port Binding, Access Control Lists (ACL), DoS protection, 802.1X, Radius Authentication, DHCP Snooping, and more. These security measures help protect the network from unauthorized access and attacks, which is important in environments with sensitive information Management Options: Integrated into Omada SDN for centralized management, as well as standalone management through Web UI, CLI, SNMP, RMON, and dual image support, making it versatile and easy to control Quality of Service (QoS): Supports L2/L3/L4 QoS and IGMP snooping to optimize voice and video applications, ensuring efficient use of bandwidth for critical services			
Patch panels  Figure 3.2.3	D-Link NPP-C61BLK24 (24-port CAT6) Category: Cat6, supporting up to 10 Gbps network speeds, suitable for most modern networking needs and helps organize Ethernet cabling in a structured and easily maintainable format Port Count: 24 RJ45 ports, it is suitable for multiple connections across labs, faculty offices, and shared spaces, providing high-speed data transfer rates essential for academic and administrative applications.	178.88	6	1073.28

	Mounting: Standard 19-inch rack-mountable design, ideal for server racks and network cabinets Durability: Constructed from high-quality metal with a solid build that ensures longevity in high-usage environments Cable Management: Integrated rear cable management features help maintain neat, organized cabling and prevent tangling			
Ethernet cables  Figure 3.2.4	UGREEN Cat6 Ethernet Cable Data Speed: Transmits at speeds up to 1000 Mbps (1 Gbps), making it ideal for high-speed applications like cloud computing and HD video streaming. Bandwidth: Provides a 250 MHz frequency, ensuring better stability compared to Cat5 and Cat5e cables. Durability: Equipped with a flexible PVC outer jacket for protection and long-lasting use. Connectors: Features gold-plated RJ45 connectors, ensuring corrosion resistance and optimal data transfer performance. Ease of Installation: Its design is ideal for easy installation in office or institutional environments.	9.85/2 m	250 m	2462.5
Wireless access point	Tp-Link AX3600-EAP660 HD Ultra-Fast Wi-Fi 6 Speeds: 1148Mbps on 2.4GHz and 2402Mbps on 5GHz Wireless Standards: IEEE 802.11ax, also known as Wi-Fi 6 ensuring capacity, performance and efficiency	1758.00	10	17580

 Figure 3.2.5	Frequency Band Class: Dual band, where connection can be done through two different frequencies which aid in overall network performance High-Density Connectivity: 4x increased capacity, meaning higher throughput median in dense environment compared to 4x4 11ac access point Centralized Management: Omada Software Defined Networking platform, singularizing all access points configuration into one, also enhancing security, efficiency and reliability Seamless Roaming: Unaffected video streams as users move between floors, maintaining a consistent connection without buffering or disruptions Placement: Wall / Ceiling mount Feature: Zero-Touch Provisioning (ZTP)[§] to reduce manual intervention, allowing automatic configuration and deployment of networking devices efficiently			
Fiber optics  Figure 3.2.6	OM4-LC/UPC-LC/UPC-DX-1M Fiber Optic Fiber Mode: OM4 50/125μm, providing a higher bandwidth up to 4700 MHz/km bandwidth at a wavelength of 850 nm. Also allow longer reach of up to 550meters at 10Gbps, and up to 150 meters for 40 Gbps and 100 Gbps transmissions. This ensures high data transmission between short to medium range.	20.00/1 m	486.5m	9730

	Fiber Count: 2 Fibers, ideal for simplex or duplex links, providing reliable high data transmission Type: LC UPC Duplex, allowing high-density applications while minimizing signal reflection for a better network connection performance Cable Type: Tight Buffered, protecting every individual fibre from physical stress Jacket Rating: OFNR (Riser), flame retardant materials to reduce fire widespread along the cable Wavelength: 850/1300nm, versatile for different distance and ensuring best data transmission performance Temperature Range: -20 to 70°C			
Total:				RM886 95.78

3.3 Reflections

3.3.1 Are you surprised by the prices? How were you surprised?

Yes, the cost of networking equipment corresponds to its advanced features and technological capabilities. After discussions with the team, we finalized the quantities needed for each networking device, which resulted in a total cost of **RM88,695.78**. This amount constitutes approximately 10% of our overall budget. However, this estimate excludes the costs associated with the power station, building maintenance, furniture, decorations, and unforeseen expenses. Some devices, such as the TP-Link TL-SG3428X switch at RM1,445 and the Tp-Link AX3600-EAP660 HD wireless access point at RM1,758, appear to be reasonably priced, whilst others, such as the TP-Link Archer AX90 router at RM799.

What surprises us is how intricate and advanced technology these devices include. The switch includes advanced security features such as IP-MAC-Port Binding and DoS prevention, the router includes tri-band Wi-Fi 6 with a 1.5 GHz quad-core processor, and the wireless access point provides lightning-fast Wi-Fi 6 speeds with centralized management. We believe some hidden variables may also influence the price such as a brand's reputation, driving them to achieve higher prices in the market. While the price may seem high at first, they represent the complex networking devices needed to operate efficiently and effectively in an IT building, making the prices seems more justifiable. Some networking devices also provide with better security measures, for example FS OM4-LC/UPC-LC/UPC-DX-1M Fiber Optic's ability to prevent fire widespread through the cables. These features suggest cost-efficient methods to ensure the building durability for the next 20 years.

3.3.2 Have you ever considered cost as a factor for choosing networking devices?

Yes, when selecting networking equipment, price is an important consideration. We prior technical standards, compatibility, and dependability of the devices but it is crucial to balance them with financial limitations. For instance, certain high-end devices could include advanced features that are unnecessary for the requirements of a particular network, thus a less expensive alternative is more suitable and cost-effective.

In our project, cost considerations include the initial cost of the device, ongoing costs, scalability, and the trade-off between cost and performance. We must be able to afford the initial purchase price, but we also need to consider long-term expenses like longevity, maintenance, and energy efficiency since equipment with greater initial prices may end up saving money over time. A crucial component is scalability, so more costly devices with greater capacity may accommodate future network expansions without requiring urgent updates. Finally, brand and warranty are important since, even if they are more expensive, premium brands often provide superior warranties and assistance, reducing the need for unforeseen repairs or replacements.

3.3.3 What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

1) Router

For the router used in this project, we have chosen the TP-Link Archer AX90 (AX6600) router. The major difference between TP-Link Archer AX90 (AX6600) with other different brands we can compare is **Asus RT-AX86U**.

Specification	Asus RT-AX86U	TP-Link Archer AX90 (AX6600)
Max Wi-Fi Speed	Up to 5700 Mbps (Dual-Band)	Up to 6600 Mbps (Tri-Band)
Feature	Dual-Band, Advanced QoS, AiMesh	Tri-Band, HomeShield, 2.5Gbps Port
Security	Lifetime free ASUS AiProtection Pro, powered by Trend Micro™	HomeShield Basic
Price (RM)	1409	799.00

The Asus RT-AX86U and TP-Link Archer AX90 (AX6600) have varied features that are appropriate for varying budgets and purposes. The TP-Link Archer AX90 offers up to 6600 Mbps with a tri-band configuration, adding another band for better performance with numerous devices, while the Asus RT-AX86U has a maximum WiFi speed of up to 5700 Mbps with a dual-band arrangement. The Asus is perfect for maximum customisation and coverage extension because of its features, which include Advanced QoS and AiMesh for mesh networking capabilities. In contrast, the TP-Link offers a 2.5Gbps WAN port and HomeShield to meet the needs of consumers who want fast internet connections with minimal security measures.

The Asus router comes with a strong, integrated security feature called AiProtection Pro, which is powered by Trend Micro and is free for life. In comparison to Asus's product, TP-Link's HomeShield Basic provides basic security but less sophisticated safeguards. Finally, the TP-Link Archer AX90 is a more cost-effective choice, coming in at around RM799.

2) Switches

For the switches used in this project, we have chosen the TP-Link TL-SG3428X switches. The major difference between TP-Link TL-SG3428X with other different brands we can compare is **D-Link DGS-1210-28MP**.

Specification	D-Link DGS-1210-28MP	TP-Link TL-SG3428X
Ports	24 x Gigabit PoE, 4 x GbE/SFP combo	24 x Gigabit RJ45, 4 x 10G SFP+
Uplink Capability	4 x GbE/SFP combo ports	4 x 10G SFP+ for high-speed uplinks
Switching Features	L2+ switching, static routing	L2+ switching, L3 QoS, IGMP snooping
Security Features	Advanced L2 security features	IP-MAC-Port Binding, ACL, DoS, 802.1X, RADIUS
Price	2284	1445

Both the D-Link DGS-1210-28MP and the TP-Link TL-SG3428X are Layer 2+ switches that have unique characteristics to meet various networking requirements. The D-Link model is perfect for networks that need to provide power to devices like IP cameras and access points since it has 24 Gigabit PoE ports, 4 GbE/SFP combination connections, and a substantial 370W PoE budget. Additionally, it incorporates L2+ switching with static routing for fundamental network security and segmentation.

The TP-Link TL-SG3428X, on the other hand, has four fast 10G SFP+ uplink ports and 24 gigabit RJ45 ports, which increase bandwidth capacity for networks with massive data traffic. It also has advanced safeguards that improve network safety for critical situations, such as Radius Authentication, DoS prevention, and IP-MAC-Port Binding. The TP-Link model, which costs around RM1445, is more affordable than the D-Link, which costs RM2284.

3) Patch panel

For the patch panel used in this project, we have chosen the D-Link NPP-C61BLK24 (24-port CAT6) patch panel. The major difference between D-Link NPP-C61BLK24 (24-port CAT6) with other different brands we can compare is **Panduit CPP24FMWBLY 24-Port CAT6 Patch Panel**.

Specification	Panduit CPP24FMWBLY 24-Port CAT6 Patch Panel	D-Link NPP-C61BLK24 (24-port CAT6)
Port Count	24 x CAT6 ports	24 x CAT6 ports
Construction	Heavy-duty construction for industrial use	Durable metal body
Labeling and Identification	Enhanced cable management options	Basic cable management
Labeling and Identification	Clear labelling with color-coded port icons	Numbered ports for easy organization
Target Environment	Enterprise-grade, suitable for high-density networks	Small to medium-sized setups
Price	368.36	178.88

Panduit Patch Panel and the D-Link Patch Panel both include 24 CAT6 ports to meet various networking requirements. The Panduit panel is designed for enterprise-level applications, with heavy-duty construction, improved cable management options, and colour-coded markings for simple port identification, making it perfect for high-density configurations. The D-Link panel, on the other hand, is ideal for small to medium-sized networks, with a sturdy metal shell, basic cable organisation, and straightforwardly numbered ports for simple organisation. Within our project's given budget, the D-Link NPP-C61BLK24 is a suitable choice as it provides a practical and economical solution.

4) Ethernet cables

For the ethernet cables used in this project, we have chosen the URGREEN Cat6 Ethernet Cable. The major difference between URGREEN Cat6 Ethernet Cable with other different brands we can compare is **D-Link Cat6 Ethernet Cable**.

Specification	D-Link Cat6 Ethernet Cable	URGREEN Cat6 Ethernet Cable
Data Speed	Up to 1000 Mbps	Up to 1000 Mbps
Bandwidth	250 MHz	250 MHz
Connectors	Standard RJ45 connectors (may not always be gold-plated)	Gold-plated RJ45 connectors for corrosion resistance
Durability	Similar durable PVC outer jacket	Flexible PVC outer jacket for protection and long-lasting use
Additional Features	Standard performance for general-purpose networking	Enhanced stability for HD streaming and cloud computing
Price	20.00 for 2 meters	9.85 for 2 meters

Both the URGREEN and D-Link Cat6 Ethernet Cables have a bandwidth of 250 MHz and support data rates of up to 1000 Mbps. The D-Link cable is designed for general-purpose networking, featuring standard RJ45 connectors (which may not always be gold-plated), a durable PVC outer jacket, and reliable performance, making it a practical choice for everyday use. The URGREEN cable, on the other hand, is suitable for HD streaming and cloud computing, with gold-plated RJ45 connections that prevent corrosion, a flexible PVC outer jacket that increases longevity, and better stability for demanding applications. Finally, since the URGREEN Cat6 Ethernet Cable costs 9.85 for 2 meters, which is half as much as the D-Link cable, it is a more affordable choice.

5) Wireless access point

For the wireless access point used in this project, we have chosen the Tp-Link AX3600-EAP660 HD. The other different brands that we can compare is **Ubiquiti UniFi U6-LR** since both utilise the same Wi-Fi standard.

Specification	Ubiquiti UniFi U6-LR	Tp-Link AX3600-EAP660 HD
Wi-Fi Standard	Wi-Fi 6 (802.11ax)	Wi-Fi 6 (802.11ax)
Max Throughput	3 Gbps (2.4 GHz: 600 Mbps, 5 GHz: 2.4 Gbps)	3.6 Gbps (2.4 GHz: 1.1 Gbps, 5 GHz: 2.4 Gbps)
Ethernet Port	1 × 1GbE	1 × 2.5GbE
Antennas	4 internal antennas	8 internal antennas
Range	Long-range coverage (ideal for large areas)	Moderate (focus on performance)
Advanced Features	MU-MIMO, Beamforming	MU-MIMO, Beamforming, Airtime Fairness
Price (RM)	1758.00	1153.00

Both wireless access point shares the same Wi-Fi standard, which is 802.11ax with latest Wi-Fi 6 quality features. However, the max throughput of Tp-Link AX3600-EAP660 HD is higher compared to Ubiquiti UniFi U6-LR, with 3.6Gbps and 3Gbps respectively. Tp-Link AX3600-EAP660 HD also has more ethernet ports which allow for link aggregation. This feature is highly significant especially in a high-density environment to increase bandwidth and throughput. Besides, Tp-Link AX3600-EAP660 HD has 4 more internal antennas, thus extending omni-directional coverage. Since the building is about medium size (40mx27m), larger coverage area is less needed compared to higher bandwidth performance, so Tp-Link AX3600-EAP660 HD is much preferred here. It also introduces new feature (Airtime Fairness) where faster client will automatically receive more bandwidth so that their individual connection speed won't be bogged down by slower clients. The only downside here is the price where Tp-Link AX3600-EAP660 HD is more expensive than Ubiquiti UniFi U6-LR, but this can be justified with its greater ethernet port numbers, antenna quantity and the extra advanced features.

6) Fiber optics

For the fiber optics used in this project, we have chosen the FS OM4-LC/UPC-LC/UPC-DX-1M Fiber Optic. The other brand model we can compare is the **Fibertronics OS2-LC/UPC-LC/UPC Duplex-3M Fiber Optic**.

Specification	OS2-LC/UPC-LC/UPC Duplex-3M Fiber Optic	OM4-LC/UPC-LC/UPC-DX-1M Fiber Optic
Brand	Fibertronics	FS
Fiber Mode	OS2 Singlemode 9/125 μ m	OM4 Multimode 50/125 μ m
Type	UC LPC Duplex	UC LPC Duplex
Jacket Type	PVC	OFNR
Electrical Interference Immunity	Yes	Yes
Pull-proof	Yes	Yes
Price (RM)	14.6/m	20/m

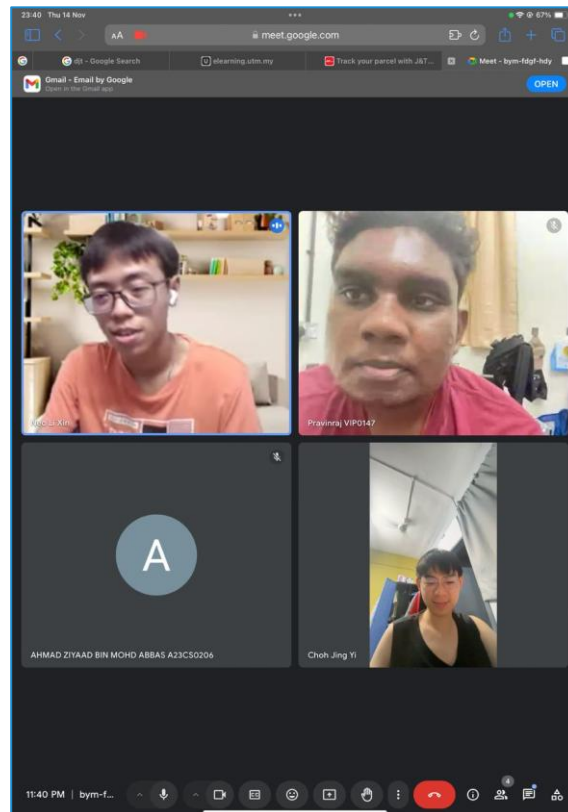
First, the biggest difference between these two fiber optics is singlemode fiber and multimode fiber. Singlemode is typically ideal for long distance while multimode for short distance. Besides, singlemode has a higher bandwidth while multimode has a higher data rate. However, we chose multimode fiber optics for our building as it meets the specific requirements of our medium size building. For the bandwidth aspect, we could improve its performance with the assistance of routers and wireless access point. On the other hand, they each have different jacket type. OFNR is much more fire resistant and non-conductive, thus potentially lowering maintenance costs associated with fire hazards and damage repairs. Although FS OM4-LC/UPC-LC/UPC-DX-1M Fiber Optic has its downsides from performance and price aspect, we still choose it due to it providing higher safety measures to accommodate the building for the next 20 years.

Meeting Minutes

Meeting Minutes #1

Date/Time		14/11 November 2024 11:00pm	
Location Online		Google Meet	
Agenda		1. Overviewing task 3 2. List the network devices we want to chose 3. Distribute task among members	
Meeting MC		Neo Li Xin	
Attendance			
Name		Time	Reason For Absence
Choh Jing Yi		23 00	-
Neo Li Xin		23 00	-
Pravinraj a/l Sivabathi		23 00	-
Ahmad Ziyaad bin Mohd Abbas		23 00	-
Meeting details			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Overviewing task 3 and make a simple discussion	-Neo Li Xin share the task 3 information and rubric -All members read the task clearly and try to understand it	All members
2	Suggest the network devices we can use in the building	-In the task, there have list that routers, switches, patch panels, wireless devices, and cables. So, Choh Jing Yi and Neo Li Xin suggest another network devices we can use based on the task 1 floor planning.	Choh Jing Yi and Neo Li Xin
3	Task distribution for each group members	-Choh Jing Yi distribute task to each member equally. -Choh Jing Yi do routers, switches, patch panels and cables	All members

		-Neo Li Xin do wireless access point and fiber optics.	
4	Meeting ended	We end our meeting at 11.30pm after all the discussion has been done.	All members



Meeting 1 Picture

Meeting Minutes #2

Date/Time		19/11 November 2024 10:00pm	
Location Online		Google Meet	
Agenda		1. Review the current progress of each member. 2. Check if the suggested product for comparison is suitable. 3. Decide the quantity and calculate the total price of each device. 4. Check the reference format to see if it matches.	
Meeting MC		Choh Jing Yi	
Attendance			
Name	Time		Reason For Absence
Choh Jing Yi	22 00		-
Neo Li Xin	22 00		-
Pravinraj a/l Sivabathi	22 00		-
Ahmad Ziyaad bin Mohd Abbas	22 00		-
Meeting details			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Review the current progress of each group member	Choh Jing Yi share the part we do together in the word document	All members
2	Check and suggest the network devices that is suitable as an example	-All members review the network devices that proposed - Choh Jing Yi evaluate the features of the selected devices to ensure they are cost-effective and have the durability to last over the long term. -Neo Li Xin suggest the brand of network devices should be reliable and long warranty -Choh Jing Yi suggest the quantity of each network	All members

		devices based on the requirement -Choh Jing Yi check the comparison of the product suggested is considerable	
3	Calculate the total price of each network devices	-Neo Li Xin calculate the total price and make sure it is lower budget and cost effective -Choh Jing Yi leads a discussion to finalize the decision on the most suitable network devices.	Choh Jing Yi,Neo Li Xin
4	Check the reference format to see if it matches.	- Neo Li Xin check the references and make sure each reference is labelled with a number and arranged in order	Neo Li Xin
5	Meeting ended	We end our meeting at 11.30pm after all the discussion has been done.	All members

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